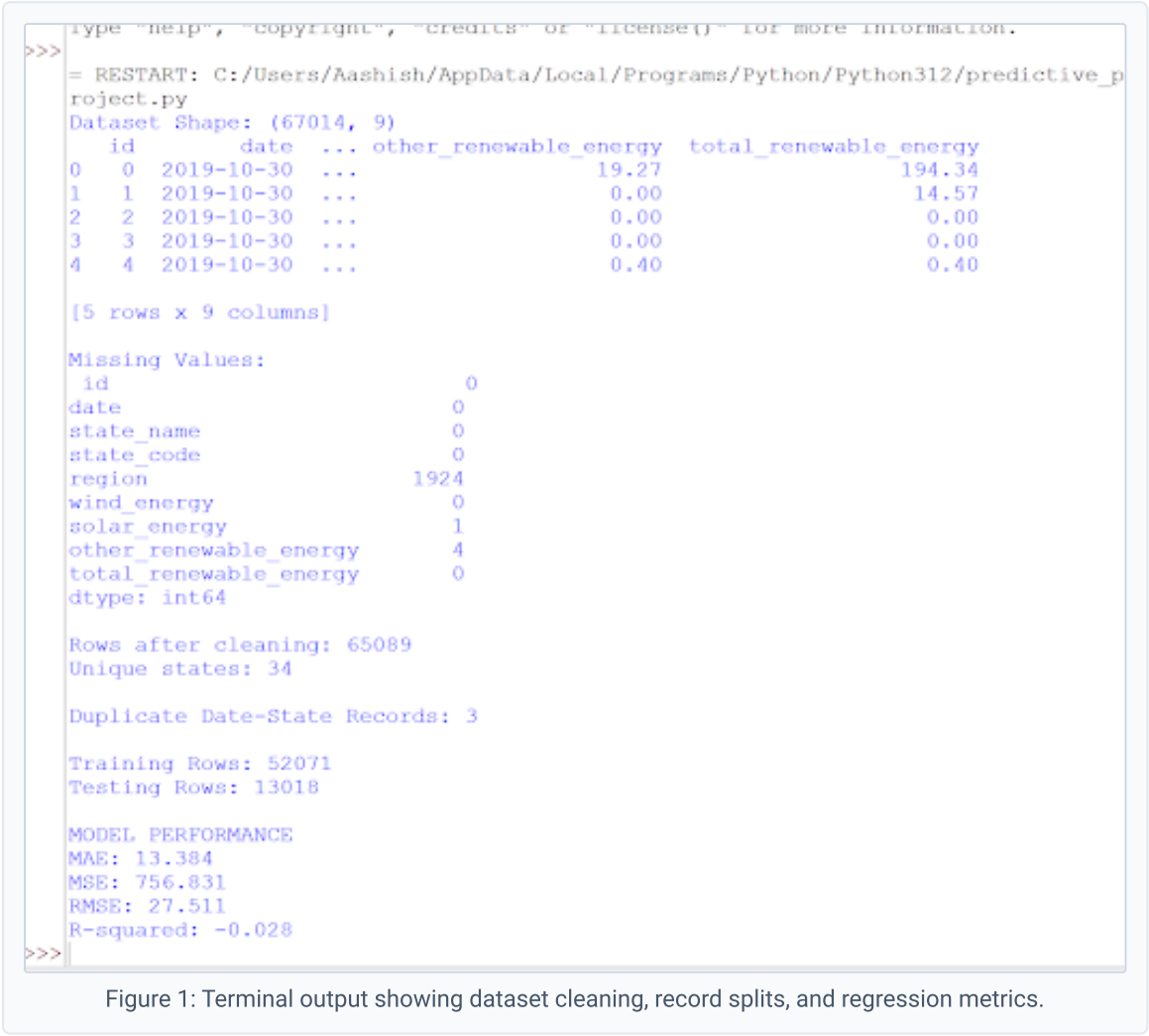
Predictive Analytics & Renewable Energy Report

Solar Energy Modeling, Data Cleaning, and Correlation Analysis

1. Model Output & Pipeline Execution

Execution log detailing dataset dimensions, data cleaning results, missing value counts, train-test splits, and baseline performance evaluation metrics.



```
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Aashish/AppData/Local/Programs/Python/Python312/predictive_p
roject.py
Dataset Shape: (67014, 9)
   id        date  ... other_renewable_energy  total_renewable_energy
0   0  2019-10-30  ...                  19.27                  194.34
1   1  2019-10-30  ...                   0.00                   14.57
2   2  2019-10-30  ...                   0.00                    0.00
3   3  2019-10-30  ...                   0.00                    0.00
4   4  2019-10-30  ...                   0.40                    0.40

[5 rows x 9 columns]

Missing Values:
 id                          0
date                        0
state_name                  0
state_code                  0
region                   1924
wind_energy                 0
solar_energy                1
other_renewable_energy      4
total_renewable_energy      0
dtype: int64

Rows after cleaning: 65089
Unique states: 34

Duplicate Date-State Records: 3

Training Rows: 52071
Testing Rows: 13018

MODEL PERFORMANCE
MAE: 13.384
MSE: 756.831
RMSE: 27.511
R-squared: -0.028
>>>
```

Figure 1: Terminal output showing dataset cleaning, record splits, and regression metrics.

2. Exploratory Data Analysis & Energy Distribution

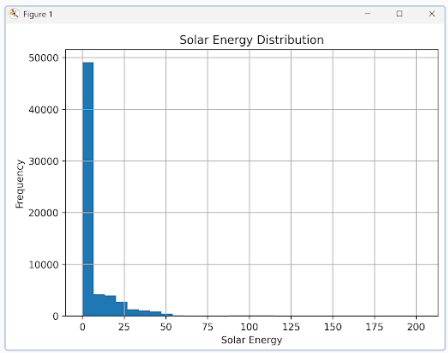


Figure 2: Distribution of Solar Energy values.

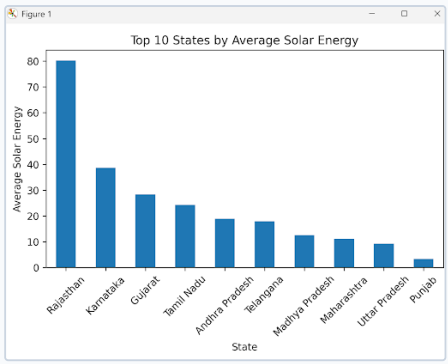


Figure 3: Top 10 Indian states ranked by average solar generation.

3. Correlation Matrix & Time Series Generation

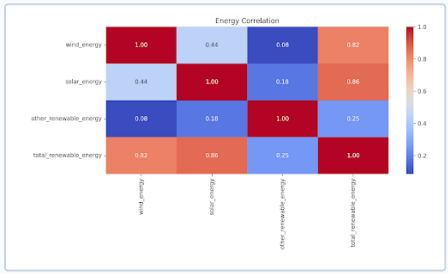


Figure 4: Correlation heatmap across renewable energy sources.

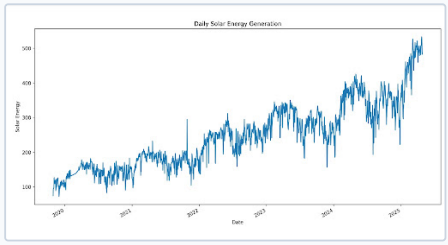


Figure 5: Daily solar energy generation timeline (2020–2025).

4. Model Evaluation: Actual vs. Predicted Values

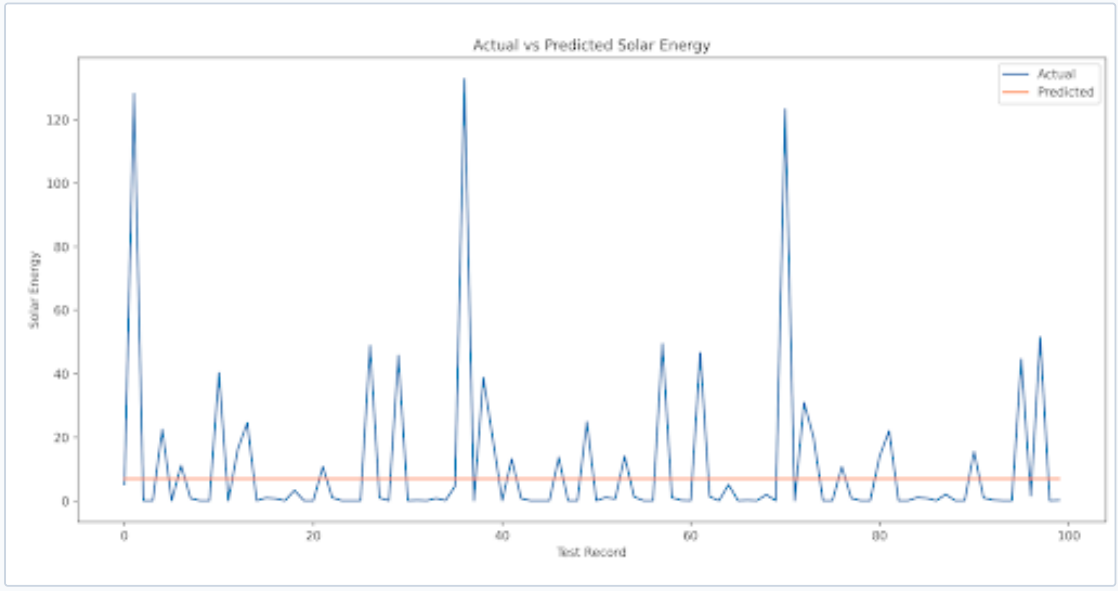


Figure 6: Model prediction overlay against actual test record benchmarks.